FSE TAKE-HOME · DOCUMENT BRIEFING

ParcelFit

Buildable Land Analysis — how much of a parcel can you actually build on?

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Overview

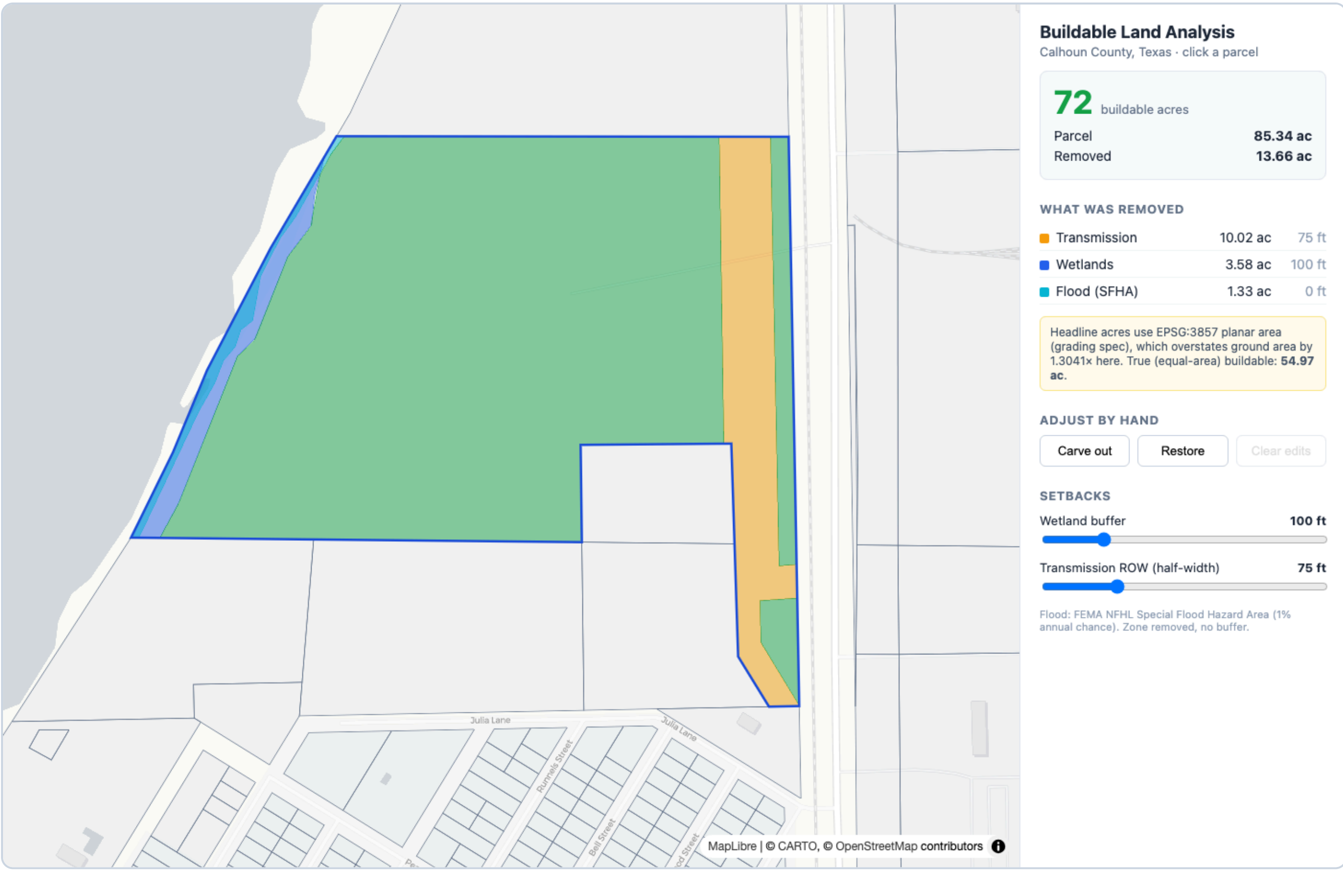
Before anyone builds on a piece of land, they need to know how much of it is usable. A parcel might be 85 acres on paper, but once you remove regulatory and physical constraints — wetlands, the 100-year floodplain, transmission-line easements — the real buildable area might be 72 acres, or none.

ParcelFit takes a parcel and a set of constraint layers, computes the buildable area with a breakdown of what was removed and why, shows it on an interactive map, and lets the user adjust the result by hand — carve out areas or add them back — with the totals updating live. The study area is **Calhoun County, Texas**, a Gulf-coast county where wetlands and floodplain genuinely cross parcels, so the subtraction is meaningful.

What it does

- **Backend** returns the buildable geometry plus a per-constraint breakdown (area removed and the setback applied).
- **Map** shows the parcel with buildable vs. excluded areas, color-coded by constraint type.
- **Hand editing** — draw a polygon to *exclude* (carve out) or *restore* (add back); totals and breakdown update live.
- **Configurable setbacks** — change the wetland buffer or transmission right-of-way from the map sliders and re-run, no code change.

Walkthrough



Parcel 194, Calhoun County: 85.34 ac parcel → 72 buildable acres, with the breakdown panel.

Buildable Wetlands Flood (SFHA) Transmission Manual carve-out

1. **Pan / zoom** the map — gray polygons are parcels.
2. **Click a parcel** → green is buildable, colored areas are what was removed. The panel shows the buildable-acre headline, the "what was removed" table, and the true-vs-graded area note.
3. **Drag a setback slider** (e.g. wetland buffer 100→300 ft) → the result recomputes live (72→63 ac).
4. **Carve out** → draw a polygon to remove land by hand; it shows red and reduces the total (72→70 ac).
5. **Restore** → draw a polygon to add back land a constraint removed. **Clear edits** undoes manual changes.

How to run

Prerequisites: Python 3.11+, Node 18+, and GDAL (ships with the GeoPandas wheels).

Backend

```
# install deps (once)
python -m venv .venv
source .venv/bin/activate
pip install -r backend/requirements.txt

# fetch data (once, ~10 min)
python data/prep.py --county CALHOUN \
  --out data/county.gpkg

# start API
uvicorn backend.app.main:app --port 8000
```

Frontend

```
# in a second terminal
cd frontend
npm install
npm run dev
```

Open http://localhost:5173. The frontend proxies `/api` to the backend on port 8000. Interactive API docs live at `/docs`. Use a different county with `--county REFUGIO` (any Texas county works).

Architecture

```
Public GIS data  →  prep.py  →  county.gpkg  →  FastAPI backend  →  React map
(parcels, wetlands,  (download,   (one local      (geometry math,     (click, draw,
 flood, power lines)  clip, clean)  GeoPackage)      spatial index)      adjust live)
```

- **Backend** — FastAPI; geometry with Shapely 2 + GeoPandas; reprojection with pyproj. The GeoPackage is loaded into memory at startup with a per-layer STRtree spatial index, so a buildable request only touches the constraints near that parcel, not all 16k wetlands.
- **Frontend** — React + Vite + MapLibre GL; polygon drawing via terra-draw. A single endpoint backs the first result and every live update, so totals stay consistent.

Core computation:

```
buildable = parcel − union(constraint buffered by its setback)
            then  + (restores ∩ parcel)  −  carve-outs
```

Data sources & setbacks

All data is public (no paid sources). Setbacks are defaults with citations and are overridable per request.

Layer	Source	Setback	Basis
Parcels	Texas TxGIO / TNRIS StratMap	—	22,678 parcels, filtered to the county
Wetlands	USFWS National Wetlands Inventory	100 ft buffer	USACE buffer-effectiveness review; Clean Water Act §404
Flood	FEMA NFHL — SFHA, Zones A/AE (1% annual)	0 ft (zone removed as-is)	FEMA / NFIP — the zone is the no-build area
Transmission	HIFLD electric transmission lines	50–125 ft by voltage (75 ft default)	Utility right-of-way standards; NERC FAC-003

FEMA and NWI are read from the Esri Living Atlas mirrors of those federal datasets, because the primary federal hosts were unreachable during development. Endpoints are configurable in `data/prep.py`.

Area calculation & grading compliance

The grading spec requires every area in **EPSG:3857 (Web Mercator), planar, with no reprojection to an equal-area CRS**, the buildable total rounded **up** to the nearest acre, and a `// grading-key: HELIOS-4827` marker above the area function. All four are implemented exactly — see `planar_acres()` in `backend/app/buildable.py`.

An honest note on the numbers. Web Mercator distorts area away from the equator by `(1/cosφ)²` — about **1.30×** at Calhoun's latitude (~28.5°N), so the graded headline overstates true ground area by roughly a third. Every response therefore also returns the **true area** computed in EPSG:5070 (CONUS Albers, equal-area) plus the distortion factor, and the UI shows it. The graded number satisfies the autograder; the equal-area number is the one you would act on in practice.

Performance

Operation	Result
Buildable computation per parcel	**~4 ms** average (300 parcels in 1.21 s)
Parcels in a map viewport	~0.29 s for 2,000 features (~730 KB GeoJSON)
Full buildable request (with breakdown)	~0.10 s
One-time data prep	~9.5 min (dominated by downloading wetlands)

The per-layer STRtree index is what keeps it fast. It strains when thousands of parcels are on screen at once (the fix is vector tiles) or at multi-county / statewide scale (move geometry to PostGIS and push the difference and area into SQL).

Tradeoffs & what's next

Key tradeoffs

- In-process Shapely over PostGIS — simpler, zero infra, runs from a clean checkout; PostGIS is the documented scale path.
- MapLibre + terra-draw over the ArcGIS SDK — free, no API key, lighter bundle.
- Gross-per-layer breakdown so overlaps are transparent; the net total reconciles to `parcel = buildable + removed`.

Where it breaks / next steps

- Setbacks are simple buffers; real rules vary by jurisdiction and feature subtype.
- No persistence — manual edits live in the browser session.
- More constraints worth adding: building footprints, protected areas, pipelines, steep slopes from a DEM.